

Sydney Girls High School



MATHEMATICS EXTENSION 1

Half Yearly

Year 11 2016

60 minutes + 5 minutes reading time

Topics: Basic Arithmetic and Algebra, Functions, Trigonometry.

Total: 60 marks

Instructions:

- There are FOUR (4) questions which are of equal value
- Attempt all questions
- Show all necessary working. Marks may be deducted for badly arranged work or incomplete working.
- Start each question on a new page.
- Write on one side of the paper only.
- Diagrams are NOT to scale.
- Board approved calculators may be used.

Name:.....

Teacher:

Question 1 (Start a new page)**Marks**

- (a) Find a and b if $\frac{2}{5-\sqrt{3}} - \frac{4}{6-2\sqrt{3}} = a + b\sqrt{3}$. 3
- (b) Solve $x^2 - 6x + 8 > 0$ and graph your solution on a number line. 3
- (c) Factorise $x^4 + x^2 - y^2 - y^4$ fully. 2
- (d) If $\sin \theta = \frac{7}{10}$ find the exact value of $\tan \theta$ if θ is obtuse. 2
- (e) Find the bearing of B from A if the bearing from B to A is 213° . 1
- (f) A ladder 2m long reaches 1.5m up a vertical wall. Find the angle the ladder makes with the ground correct to the nearest degree. 2
- (g) $x = \frac{3}{2}$ is a solution of $ax^2 + 5x - 12 = 0$. Find the other solution. 2

Question 2 (Start a new page)**Marks**

(a) Evaluate $4^{3^{2^1}}$.

1

(b) Expand and simplify $(2x - 3y)^4$.

2

(c) Solve $\frac{2}{x-1} \geq 3$.

3

(d) A ship sails on a bearing of 136° for a distance of 256km. How far South is the ship of the starting point?

2

(e) Draw the graph of $y = \cot x$ for $0^\circ \leq x \leq 360^\circ$.

2

(f) Find the exact value of

(i) $\sin 45^\circ$

1

(ii) $\cos 240^\circ$

1

(iii) $\operatorname{cosec} 300^\circ$

1

(g) Simplify

(i) $\sec \theta \cot \theta$

1

(ii) $1 + \sin(-\theta) \cos(90^\circ - \theta)$

1

Question 3 (Start a new page)**Marks**

(a) Solve $\frac{x+1}{x^2+5x+6} + \frac{x-1}{x^2-9} = 0$.

3

(b) For $y = \frac{x-4}{x+1}$

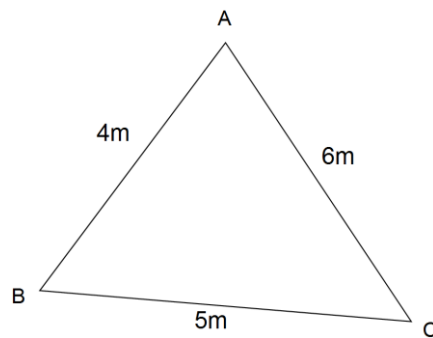
(i) Find the equations of the asymptotes.

2

(ii) Find the intercepts and sketch the curve.

2

(c)



Find the area of the above triangle correct to 2 significant figures.

3

(d) On a number plane sketch $y = 4 - |x + 3|$ showing all important features.

3

(e) Find the domain and range of $y = \frac{4}{\sqrt{1-x}}$.

2

Question 4 (Start a new page)**Marks**

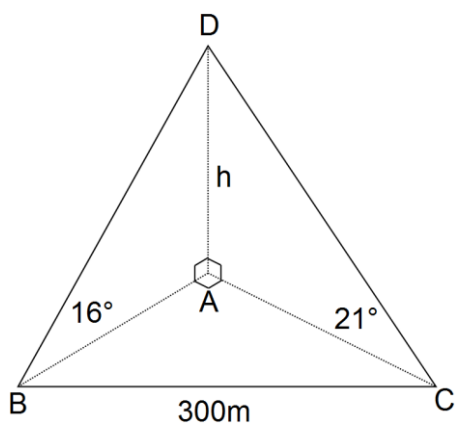
(a) (i) Graph $y = (3-x)(x-1)(x+2)$ showing the intercepts on both axes. 2

(ii) Solve $(3-x)(x-1)(x+2) \leq 0$. 2

(b) Solve $\sin x - 2 \sin x \cos x + 2 \cos x = 1$ for $0^\circ \leq x \leq 360^\circ$. 3

(c) Sketch a possible graph of $y = f(x)$ if $f(x) - f(-x) = 0$ and the graph passes through the points $(0,3)$ and $(-2,0)$. 2

(d)



The above diagram shows a triangular pyramid with a right angled triangle as its base.

(i) Find an expression for AB in terms of h. 1

(ii) Find h correct to 2 decimal places. 2

(e) Prove that $\frac{1 + \cot x}{\operatorname{cosec} x} - \frac{\sin x}{\tan x \cot x} = \cos x$. 3

The End

Q1 a) $\frac{2}{5-\sqrt{3}} - \frac{4}{6-2\sqrt{3}} = a + b\sqrt{3}$

LHS: $\frac{2(5+\sqrt{3})}{25-3} - \frac{4(6+2\sqrt{3})}{36-4 \times 3}$
 $\frac{10+2\sqrt{3}}{22} - \frac{(24+8\sqrt{3})}{24}$ ✓

$\frac{5+\sqrt{3}}{11} - \frac{(12+4\sqrt{3})}{12}$

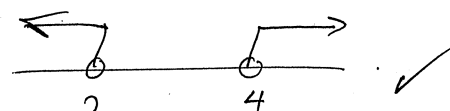
$\frac{12(5+\sqrt{3}) - 11(12+4\sqrt{3})}{132}$
 $\frac{60+12\sqrt{3} - 132 - 44\sqrt{3}}{132}$

$\frac{-72 - 32\sqrt{3}}{132} = \frac{-72}{132} - \frac{32\sqrt{3}}{132}$

Thus $a = \frac{-72}{132}$ ✓, $b = \frac{-32}{132}$ ✓ = $-\frac{8}{33}$
 $a = -6/11$

b) $x^2 - 6x + 8 > 0$

$(x-4)(x-2) > 0$ ✓

$\begin{cases} x < 2 \\ x > 4 \end{cases}$ ✓  ✓

Some students forgot to sketch the solutions on a number line as the question requested.

A number of students could not rationalise the denominators correctly

c) Factorise: $x^4 + x^2 - y^4 - y^2$

$= x^4 - y^4 + x^2 - y^2$

$= (x^2)^2 - (y^2)^2 + x^2 - y^2$

$= (x^2 + y^2)(x^2 - y^2) + (x^2 - y^2)$ ✓

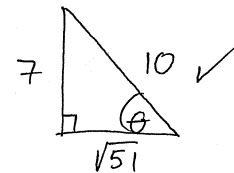
$= (x^2 - y^2)(x^2 + y^2 + 1)$

$= (x+y)(x-y)(x^2 + y^2 + 1)$ ✓

Some students did not factorise fully.

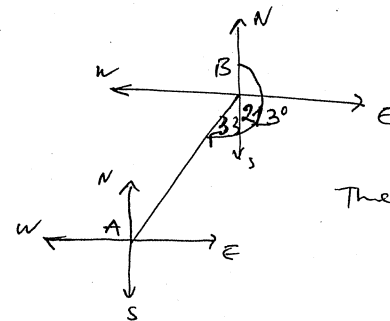
d) $\sin \theta = \frac{7}{10}$

$\tan \theta = -\frac{7}{\sqrt{51}}$ ✓



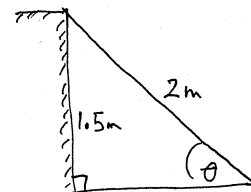
$\tan \theta < 0$ for $90^\circ < \theta < 180^\circ$

e)



The Bearing of B from A is 033° ✓

f)



$\sin \theta = \frac{1.5}{2}$ ✓ $\Rightarrow \theta = 49^\circ$ (nearest degrees) ✓

g) $ax^2 + 5x - 12 = 0$. If $x = \frac{3}{2}$

$a \times \frac{9}{4} + 5 \times \frac{3}{2} - 12 = 0$

$a = 2$ ✓

OR

$\frac{x^2}{2} + 5x - 12 = 0$
 $2x^2 + 50x - 120 = 0$

$(2x-3)(x+4) = 0$

$x = \frac{3}{2}$ ✓
 $x = -4$ ✓

Question 2

(a) $4^3 = 4^9 = 262\ 144$

(a) was incorrectly evaluated by many students. A simple way to do this would have involved using your calculator.

(b)
$$\begin{aligned}(2x-3y)^4 &= (2x)^4 - 4(2x)^3(3y)^1 + 6(2x)^2(3y)^2 \\ &\quad - 4(2x)(3y)^3 + (3y)^4 \\ &= 16x^4 - 96x^3y + 216x^2y^2 - 216xy^3 + 81y^4\end{aligned}$$

This could have also been expanded using
 $(2x-3y)^2(2x-3y)^2 = (4x^2-12xy+9y^2)(4x^2-12xy+9y^2)$
=

Students need to be careful with simplifying terms and be careful with the + or - sign.

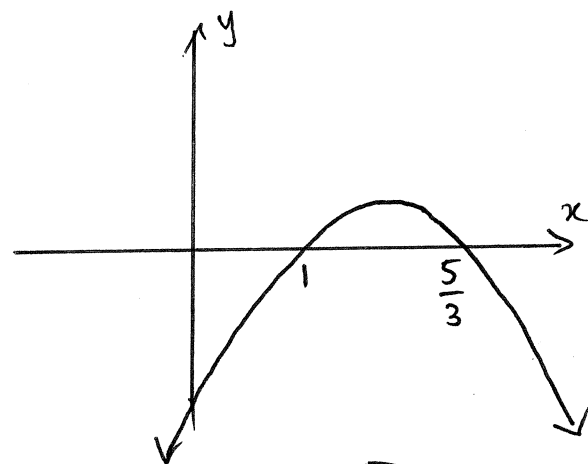
(c) $\frac{2}{x-1} x(x-1)^x \geq 3(x-1)^2$

$$(x-1)[2-3(x-1)] \geq 0$$

$$(x-1)(2-3x+3) \geq 0$$

$$(x-1)(5-3x) \geq 0$$

$$\therefore 1 \leq x \leq \frac{5}{3}$$

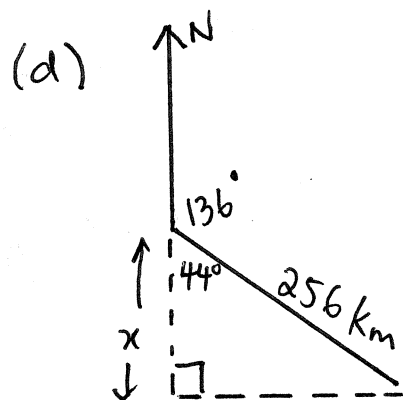


Since $x \neq 1$, the solution is $1 < x \leq \frac{5}{3}$.

Other methods were used to solve the inequation, including consideration of critical points.

Common errors included excluding $x=1$ from the solution set.

Question 2 (continued)

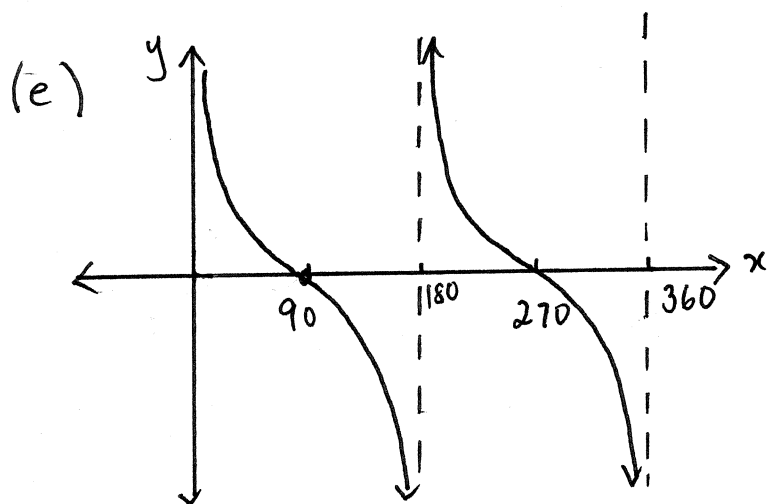


$$\cos 44 = \frac{x}{256}$$

$$x = 256 \cos 44$$

$$\therefore x \doteq 184.2 \text{ km}$$

Well done by
vast majority.



Many "attempts" were
poorly done.

Asymptotes must be
clearly marked.

The graph should be
clearly shown to be
approaching them.

(f) (i) $\sin 45 = \frac{1}{\sqrt{2}}$

(ii) $\cos 240 = \cos(180+60)$
 $= -\cos 60$
 $= -\frac{1}{2}$

S/A
T/C

(iii) $\operatorname{cosec} 300 = \frac{1}{\sin 300}$
 $= \frac{1}{-\sin 60}$
 $= -\frac{2}{\sqrt{3}}$

Since $\sin 60 = \frac{\sqrt{3}}{2}$

Mostly done
well.

However, many
students did
not recognise
the - sign
in (ii) and (iii)

(g) (i) $\sec \theta \cot \theta = \frac{1}{\cos \theta} \times \frac{\cos \theta}{\sin \theta} = \frac{1}{\sin \theta}$ or $\operatorname{cosec} \theta$

(ii) $1 + \sin(-\theta) \cos(90-\theta) = 1 - \sin \theta \times \sin \theta$
 $= 1 - \sin^2 \theta$
 $= \cos^2 \theta$

This was one of the weaker parts of this question.
Some students need to ensure they read the solution.

$$3(a) \frac{x+1}{(x+2)(x+3)} + \frac{x-1}{(x+3)(x-3)} = 0$$

$$(x+1)(x-3) + (x-1)(x+2) = 0$$

$$x^2 - 2x - 3 + x^2 + x - 2 = 0$$

$$2x^2 - x - 5 = 0$$

$$x = \frac{1 \pm \sqrt{1^2 - 4 \times 2 \times -5}}{2 \times 2}$$

$$= \frac{1 \pm \sqrt{41}}{4}$$

$$(b)(i) y = \frac{x+1}{x+1} - \frac{5}{x+1}$$

$$= 1 - \frac{5}{x+1}$$

The equations of the asymptotes are not $x \neq -1$ and $y \neq 1$

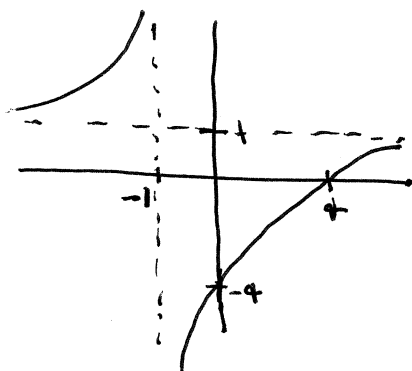
$$\therefore x = -1, y = 1$$

$$(ii) x = 0, y = -\frac{4}{1}$$

$$= -4$$

$$y = 0, x - 4 = 0$$

$$x = 4$$



Students needed to show all intercepts on their graphs.

$$(c) \cos \hat{B} = \frac{5^2 + 4^2 - 6^2}{2 \times 5 \times 4}$$

$$\hat{B} = 92.11169811$$

$$A = \frac{1}{2} \times 5 \times 4 \times \cos(92.11169811)$$

$$= 9.921567416$$

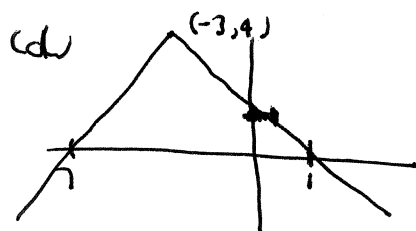
$$\approx 9.9$$

One should not round off in the middle of a calculation

$$(e) \text{ domain } 1-x > 0$$

$$-x > -1$$

$$x < 1$$



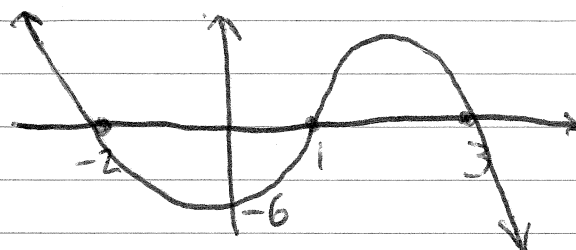
Many students left out some of the intercepts

$$\text{range } y > 0$$

206 Ext 1 Yr 11 Task 1

Q4.

(a) (i)



When $x=0$
 $y = (3)(-1)(2).$
 $= -6.$

(ii) $-2 \leq x \leq 1$, $x \geq 3$.

(b) $\sin x - 2 \sin x \cos x + 2 \cos x - 1 = 0.$

$\therefore (1 - \sin x) + 2 \cos x (1 - \sin x) = 0.$

$(1 - \sin x)(2 \cos x - 1) = 0.$

$\sin x = 1$ or $\cos x = \frac{1}{2}.$

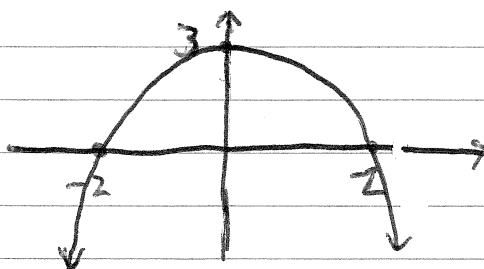
$x = 90^\circ.$

$x = 60^\circ, 300^\circ$

Factorising by
pairs is an
important technique

$\therefore x = 60^\circ, 90^\circ, 300^\circ$

(c) If $f(x) = f(-x)$ then $f(x) = f(-x)$
that is $f(x)$ is even.



y-intercept 3.
Goes through $(-2, 0)$ $(2, 0)$
and symmetrical about
y-axis.

Many Students didn't recognise this
as an even function.

$$(d)(i) \quad \tan 16^\circ = \frac{h}{AB}$$

$$AB = \frac{h}{\tan 16^\circ}$$

$$(ii) \quad AC = \frac{h}{\tan 21^\circ}$$

$$AC^2 + AB^2 = 300^2$$

$$\frac{h^2}{\tan^2 21^\circ} + \frac{h^2}{\tan^2 16^\circ} = 300^2$$

$$h^2 \left(\frac{1}{\tan^2 21^\circ} + \frac{1}{\tan^2 16^\circ} \right) = 300^2$$

$$h = \frac{300}{\sqrt{\frac{1}{\tan^2 21^\circ} + \frac{1}{\tan^2 16^\circ}}}$$

$$\approx 68.92 \text{ m}$$

$$(e) \quad \text{LHS} = \frac{1 + \frac{\cos x}{\sin x}}{\frac{1}{\sin x}} \times \frac{\sin x}{\sin x} - \frac{\sin x}{\frac{\sin x}{\cos x} \times \frac{\cos x}{\sin x}}$$

$$= \frac{\sin x + \cos x}{1} - \frac{\sin x}{1}$$

$$= \cos x$$

$$= \text{RHS.}$$

A common mistake was to square 300 and get 900.

Some students confused $\sin x + \cos x$ with $\sin^2 x + \cos^2 x$